[redacted]

From: Steve Greer <[redacted]>
Sent: Thursday, April 5, 2018 1:26 PM
To: [redacted]
Subject: Fwd: 500 bp cited in paper

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From: **red** <[redacted]u>
Date: Wed, Mar 28, 2018 at 1:10 PM
Subject: 500 bp cited in paper
To: Steve Greer <[redacted]>

Dear Steve, I was just quoting from the published paper:

"Taken together with the DNA damage results above, this indicates that the Ata DNA was relatively free of DNA damage and contaminants. Moreover, the average DNA fragment size for Ata is ~300 bp which, based on a DNA-decay model (Allentoft et al. 2012), is consistent with a sample younger than 500 yr."

I am not sure how he made this calculation, given so many caveats in the 2012 paper about different temperatures and so on?

Allentoft ME, [redacted], [redacted] et al. 2012. *The half-life of DNA in bone: measuring decay kinetics in 158 dated fossils. Proc Biol Sci 279*: 4724–4733.

The key point here is that the small mummy DNA was damaged so as to cause a double-strand break on the average once every 300 bp. Typically such degraded DNA molecules also show single-stranded nicks which are held together as part of a double helix with a size range somewhat less, maybe 100-200 bases. He did not make this easy measurement from a s.s. gel!

So the probability is that some organic base C, T, A or G might be damaged on approximately the same scale of size <300 bases. Damaging a base does not always break a sugar-phosphate chain, unless the sample is heated to 90 C at high pH for a while, or unless you add an enzyme UNG which cuts at the damaged base, so that it will not amplify by PCR.

The obvious control which most people do is to add UNG and see what happens? Does your DNA get shorter as a whole, because it has some damaged C bases? That is another thing which Nolan and colleagues did not do. If they simply had added UNG enzyme for 30 minutes at 37 C, and the **single-stranded form** of that sample got **shorter**, then the whole paper becomes worthless as an artefact! I did not wish to be so direct and insulting in a formal set of comments. A **one-hour** cheap experiment to assess whether your sequence results might be faulty due to age-damaged C bases.

If on the other hand, those "novel" sequences somehow turn out to be "real", then there are other bioinformatics methods to identify "new genes" which he also did not use. You can check to see whether such mutations are random or directed in the Genetic Code, by whether they change an amino acid? Random mutations only change an amino acid 1/3 of the time, while directed mutations (say from an alien gene) might change amino acids 100% of the time. So if all of the changes in some particular gene change various amino acids, then it is not random. If only 1/3 of them change an amino acid, it is random.

Finally it may be better for you **not** to mention my name, because they may say: "The scientist who studies crop circles, don't believe him! Has he gone completely nuts?" LOL

Best wishes, [redacted] in Sydney